CAMBRIDGE
International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

0536931681

MATHEMATICS **9709/32**

Paper 3 Pure Mathematics 3 **February/March 2026**

1 hour 50 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

© Cambridge University Press & Assessment 2026

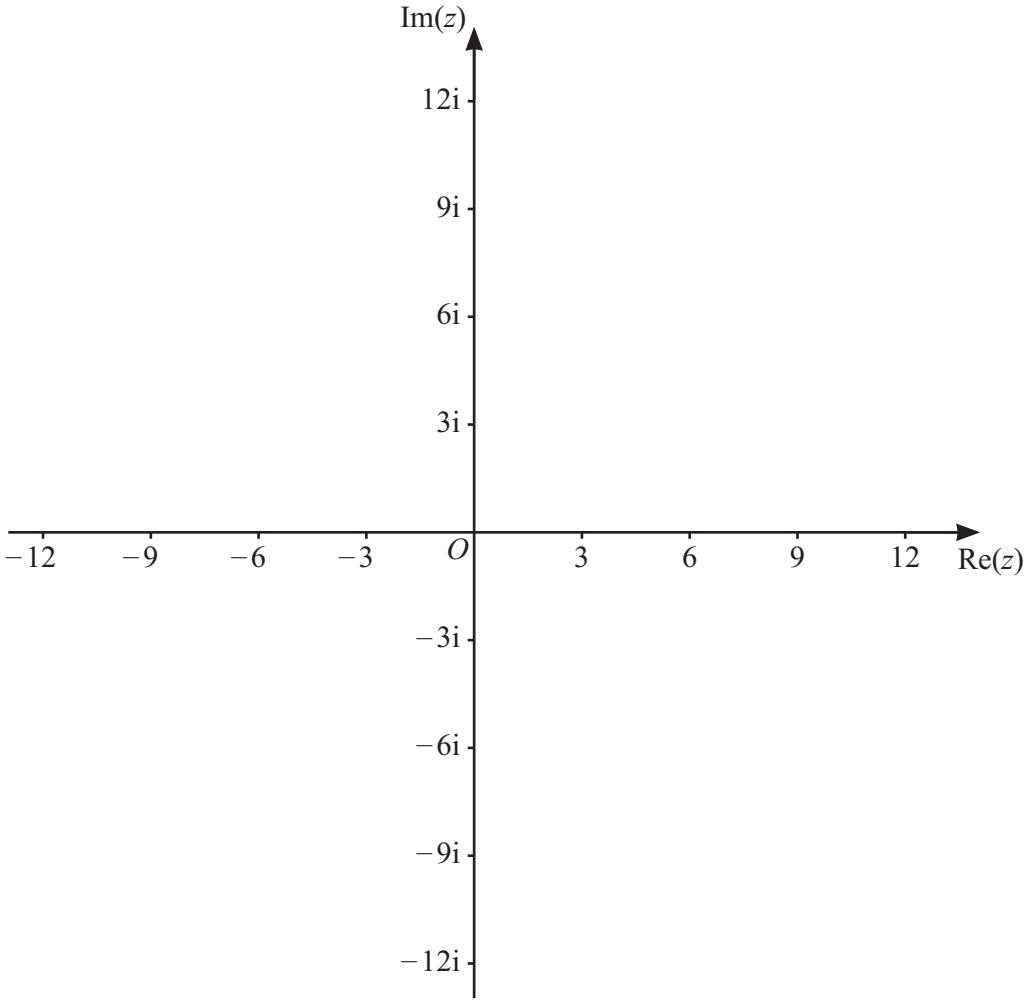
1 Solve the equation $4\times 2^{x+2} - 5\times 2^{2-x} = 3$. Give your answer correct to 3 significant figures. [4]

DO NOT WRITE IN THIS MARGIN

2 The complex number z satisfies $|z| = 9$ and $\frac{1}{2}\pi \leqslant \arg z \leqslant \pi$.

(a) On the Argand diagram, sketch the locus of the points representing z. [2]

(b) On the **same diagram**, sketch the locus of the points representing $z^* + 3$. [2]



DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

3 Find the exact value of $\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} 3\sin x \sin 2x \, \mathrm{d}x$.

Give your answer in the form $p\sqrt{3} + q\sqrt{2}$, where p and q are rational. [4]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

4 The coefficient of x^3 in the expansion of $(1-ax)^{\frac{2}{5}}$ is 1.

(a) Find the value of a. [2]

.....

.....

.....

.....

.....

.....

.....

.....

(b) Hence, find the coefficient of x^4 in the expansion of $(2x+1)(1-ax)^{\frac{2}{5}}$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) State the set of values of x for which the expansion in **4(b)** is valid. [1]

.....

.....

.....

.....

5 It is given that $z = \dfrac{3+\lambda \mathrm{i}}{\lambda+2\mathrm{i}}$, where λ is a real constant.

(a) Find the value of λ for which $\arg z = \frac{1}{4}\pi$. [4]

(b) When λ has the value found in **5(a)**, find the exact value of $|z|$, making your method clear. [2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

6 The polynomial $2x^4 + ax^3 + 4x^2 + bx - 3$ is denoted by p(x).

It is given that $\left(x^2 + x + 1\right)$ is a factor of p(x).

(a) Find the values of a and b. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(b) Hence, show that $(x + 3)$ is a factor of p(x). [2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

© Cambridge University Press & Assessment 2026

7 **(a)** By sketching a suitable pair of graphs, show that the equation $\ln x = \operatorname{cosec} \frac{1}{2}x$ has exactly one root in the interval $0 < x < \pi$. [2]

(b) Verify by calculation that this root lies between 2.6 and 2.9. [2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

© Cambridge University Press & Assessment 2026

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(c) Use the iterative formula $x_{n+1} = \exp\left(\text{cosec}\,\frac{1}{2}x_n\right)$ to determine the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

[$\exp(x)$ is an alternative notation for e^x.]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

8 The variables x and y satisfy the differential equation

$$y\mathrm{e}^{3x}\frac{\mathrm{d}y}{\mathrm{d}x} = x(y+5).$$

It is given that $y = 0$ when $x = 0$.

Solve the differential equation to obtain an equation in x and y. [8]

DO NOT WRITE IN THIS MARGIN

© Cambridge University Press & Assessment 2026

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

9 Let $I = \int_1^3 \frac{x^3}{3+x^2}\,\mathrm{d}x$.

(a) Using the substitution $x = \sqrt{3}\tan u$, show that $I = \int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} 3\tan^3 u\,\mathrm{d}u$. [4]

DO NOT WRITE IN THIS MARGIN

(b) Hence, or otherwise, find the exact value of I. Give your answer in the form $p + q \ln r$, where p, q and r are rational. [5]

DO NOT WRITE IN THIS MARGIN

10 The variables x and y satisfy the equation $y^2 = k\dfrac{x-2}{x+2}$, where k is a constant.

(a) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2y}{x^2-4}$. [5]

DO NOT WRITE IN THIS MARGIN

(b) Given that $k = 5$, find the angle between the tangents to the curve when $x = 3$.

Give your answer in the form $a\tan^{-1}\left(\frac{b}{c}\right)$, where a, b and c are integers. [4]

DO NOT WRITE IN THIS MARGIN

11 The points A and B have position vectors $\overrightarrow{OA} = 2\mathbf{i}+2\mathbf{j}-\mathbf{k}$ and $\overrightarrow{OB} = 4\mathbf{i}+2\mathbf{j}+4\mathbf{k}$ relative to the origin, O.

(a) Show that the perpendicular distance from A to the line through O and B is $\frac{1}{3}\sqrt{65}$. [5]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) The point C has position vector $\overrightarrow{OC} = 3\mathbf{i} + p\mathbf{j} + q\mathbf{k}$, where p and q are constants.

Given that

- $\overrightarrow{OC}$ is perpendicular to $\overrightarrow{AB}$ and
- angle AOC = angle COB,

find the values of p and q. [7]

Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

DO NOT WRITE IN THIS MARGIN

© Cambridge University Press & Assessment 2026



BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

© Cambridge University Press & Assessment 2026



BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (Cambridge University Press & Assessment) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in our Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge International Education is the name of our awarding body and a part of Cambridge University Press & Assessment, which is a department of the University of Cambridge.

© Cambridge University Press & Assessment 2026